

8(a)(i)	chemoreceptor ;	1
8(a)(ii)	<i>any two from:</i> 1 (different mycobacterial species produce) the same / similar, chemicals / proteins ; 2 (since) share common ancestors / same genus / share many genes / closely related ; 3 same chemoreceptors stimulated ;	2

Question	Answer		Marks														
8(b)		<table><tr><td>kingdom</td><td>Animalia</td></tr><tr><td>phylum</td><td>Chordata</td></tr><tr><td>class</td><td>Mammalia</td></tr><tr><td>order</td><td>Rodentia</td></tr><tr><td>family</td><td>Nesomyidae</td></tr><tr><td>genus</td><td><i>Cricetomys</i></td></tr><tr><td>species</td><td><i>gambianus</i></td></tr></table>	kingdom	Animalia	phylum	Chordata	class	Mammalia	order	Rodentia	family	Nesomyidae	genus	<i>Cricetomys</i>	species	<i>gambianus</i>	2
	kingdom	Animalia															
	phylum	Chordata															
	class	Mammalia															
	order	Rodentia															
	family	Nesomyidae															
	genus	<i>Cricetomys</i>															
	species	<i>gambianus</i>															
4 correct = 2 marks 1 / 2 / 3 correct = 1 mark																	

Question	Answer	Marks																																													
8(c)	any four from: comparative statements required <table><tr><td></td><td>Eukarya</td><td></td><td>Bacteria</td><td></td></tr><tr><td>1</td><td>linear DNA</td><td>and</td><td>plasmids / circular DNA</td><td>;</td></tr><tr><td>2</td><td>histones</td><td>and</td><td>no histones</td><td>;</td></tr><tr><td>3</td><td>cell walls only in some</td><td>and</td><td>cell walls in all</td><td>;</td></tr><tr><td>4</td><td>cellulose / chitin, cell walls</td><td>and</td><td>peptidoglycan cell wall</td><td>;</td></tr><tr><td>5</td><td>divide by mitosis</td><td>and</td><td>divide by binary fission</td><td>;</td></tr><tr><td>6</td><td>some sexual reproduction</td><td>and</td><td>asexual reproduction</td><td>;</td></tr><tr><td>7</td><td>(70S and) 80S ribosomes</td><td>and</td><td>70S ribosomes (only)</td><td>;</td></tr><tr><td>8</td><td>have cells > 5 μm diameter</td><td>and</td><td>cells1–5 μm diameter</td><td>;</td></tr></table>		Eukarya		Bacteria		1	linear DNA	and	plasmids / circular DNA	;	2	histones	and	no histones	;	3	cell walls only in some	and	cell walls in all	;	4	cellulose / chitin, cell walls	and	peptidoglycan cell wall	;	5	divide by mitosis	and	divide by binary fission	;	6	some sexual reproduction	and	asexual reproduction	;	7	(70S and) 80S ribosomes	and	70S ribosomes (only)	;	8	have cells > 5 μm diameter	and	cells1–5 μm diameter	;	4
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8(d)	any two from: 1 have DNA or RNA ; 2 nucleic acid is double or single stranded ; 3 AVP ; e.g.presence or absence of phospholipid envelope presence or absence of tail sheath type of host type of disease caused	2																																													